



Ideation Document for Intelligent Ground Vehicle Competition (IGVC) by Team Cognitrek from Dharmsinh Desai University , Nadiad

Team Composition :

Mentor : Prof. Gopal Gohel Gopalgohel.maths@ddu.ac.in

Team Members :

- 1) Patel Aksh Dharmeshkumar , 3rd year B.tech , Electronics and communication
24ecuos100@ddu.ac.in
 - 2) Patel Om Prakashchandra , 3rd year B.tech , Electronics and communication
24ecuof121@ddu.ac.in
 - 3) Patel Aniruddh Chetankumar , 3rd year B.tech , Electronics and communication
24ecuos101@ddu.ac.in
 - 4) Donga Sanket ParsottamBhai , 3rd year B.tech , Electronics and communication
24ecuog034@ddu.ac.in
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Mentor : Prof. Gopal Gohel	0
Team Members :	0
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1. Abstract

Team Cognitrek , Dharmsinh Desai University, presents a project for Robofest 6.0, aiming to create a highly capable autonomous rover by leveraging Raspberry Pi 5 and Pi Camera Module 3 to achieve seamless navigation and efficient task completion in challenging environments. Features include learning from the environment, autonomous driving guided by a camera, versatile mobility, and data collection. Future enhancements aim to integrate a robotic arm for sample gathering. The goal is to develop an autonomous rover with advanced AI capabilities for exploration and robotics, paving the way for enhanced scientific missions in challenging environments.

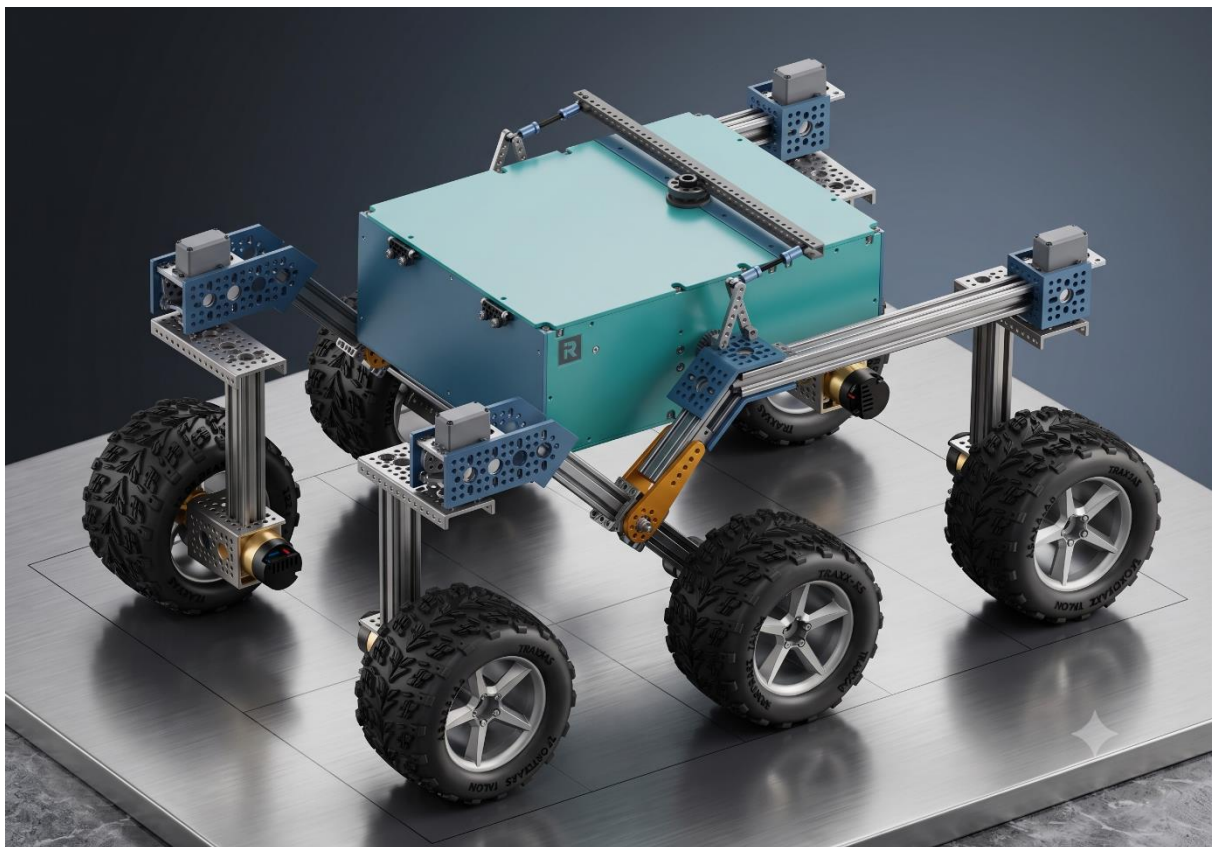


Fig (1) Proposed design of the rover

2. Features

2.1 Mechanical Features

2.1.1 Modified Rocker Bogie mechanism

- The modified Rocker-Bogie suspension system enhances the rover's ability to navigate uneven terrain by flexing and conforming to the ground's contours.

- This flexibility helps maintain stability on tilted surfaces and prevents the rover from getting stuck or toppling.
- It also ensures even weight distribution & avoids any sort of rocking.
- Our structure's unique feature is having 2 distinct rotation axes for rear & front rockings.
- Unlike the classic mechanism, the rear one rotates about an axis perpendicular to the one about which the front one rotates.
- As per our hit and trial experiments, this mechanism has proven efficient in climbing obstacles of uneven shape and traversing a rough terrain of rocks and debris (an open ground with uneven terrain).



Fig (2) Doosan DA40-5 articulated dump truck (the inspiration for our rover design)

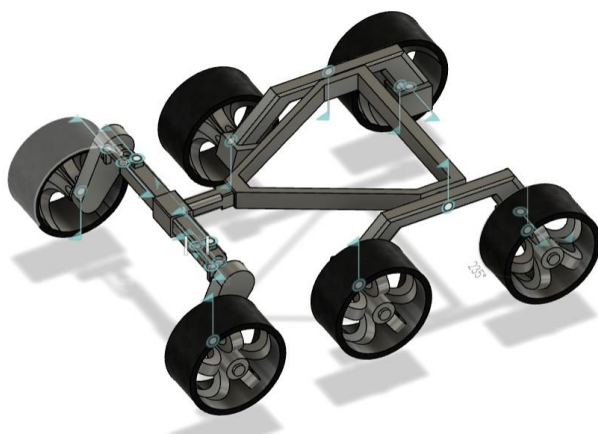


Fig (3) The chassis model

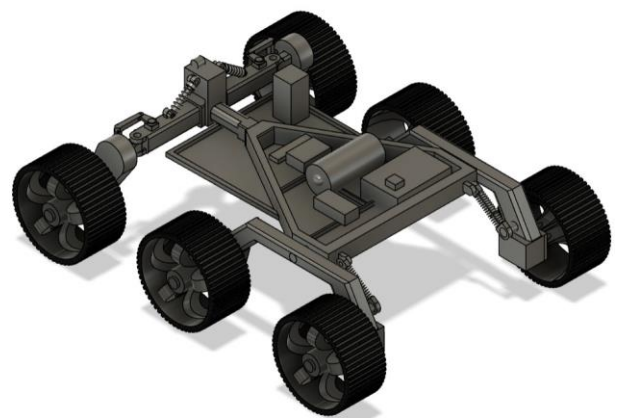


Fig (4) The final assembly

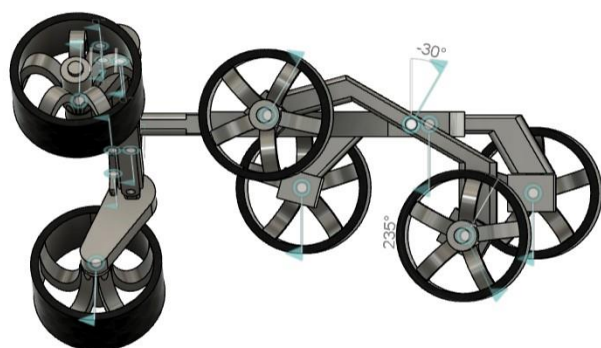


Fig (5) Left rocking (bird view)

Fig (6) Left rocking (side view)

2.1.2 Navigation mechanism

- The navigation mechanism for the rover is a 6-wheel drive with balloon-type stripped rubber casing (proposed diameter of 10-12 inches (25-30cm)).
- These 6 wheels are supposed to facilitate forward motion, backward motion & on-spot rotations with the power supplied to them by the lipo battery of 12V.
- We've also inculcated a steering mechanism for all 6 wheels, which works on the Ackermann's steering geometry. To actuate that, we've used an MG996R servo motor & a gear mechanism.
- This steering mechanism will allow the rover to move in any direction.

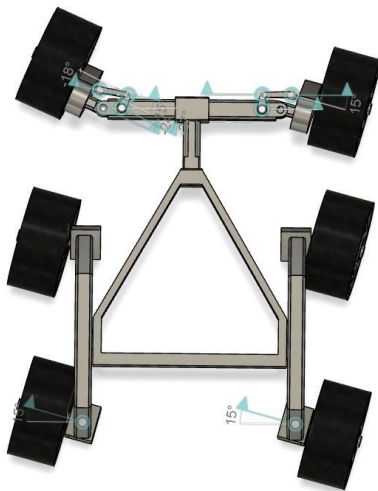


Fig (7) 15° Diagonal orientation

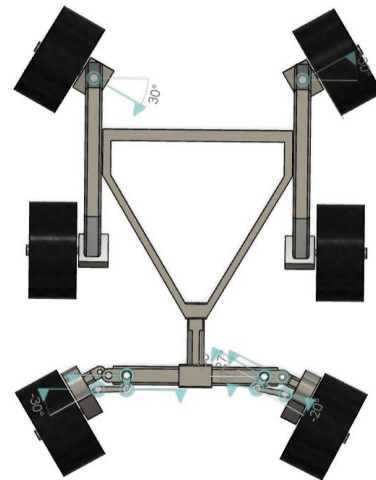


Fig (8) Orientation for In-place rotation

2.1.3 Weight reduction mechanism

- The main chassis is a casing rather than a rigid box-like structure.
- The aluminium extruders are the bones of the rover.
- We've used customised shape acrylic sheets of 5mm as "slabs" to place the components over it. Slabs are supposed to be placed in the grooves of the extrusions.
 - All the components are to be screwed over the "slab" using appropriate sort of nuts.

2.1.4 Shock Absorption Technology

- Shock-ups are installed on the axles to absorb the sudden jerks the rover may encounter due to objects in its path.
- These reduce vibrations and ensure a smoother ride over rough terrain.

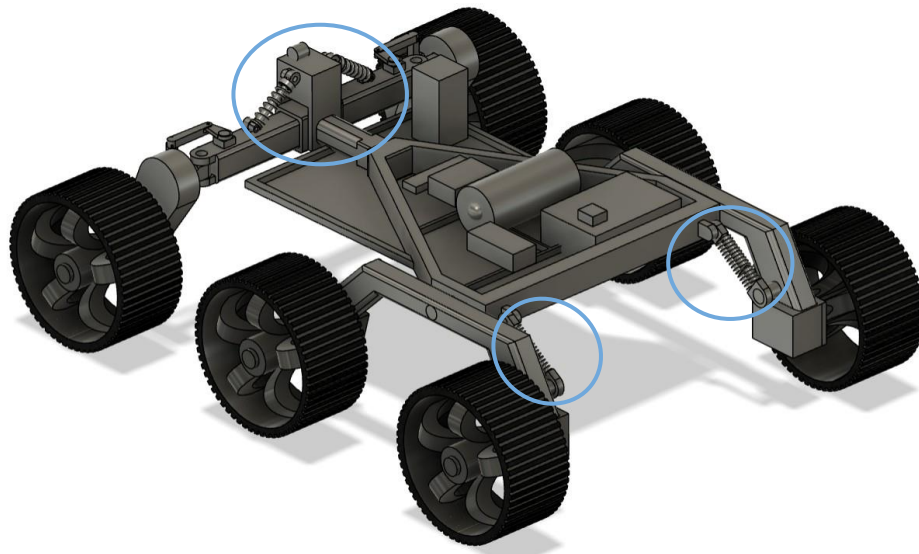


Fig (9) Shock up mechanism

2.2 Technological Features

2.2.1 Electronics

- Lithium batteries of 12V, 30Ah
- Motor driver (BTS7960 / equivalent high-current driver)
- PCA 9685
- Raspberry Pi Camera Module 3
- To retrieve RGB data from the environment
- Ultrasonic sensors
- For close-range distance in obstacles, tunnel and underpass
- Raspberry Pi 5 Board
- For data processing and decision making
- Neo-6M GPS Module
- To access GPS
- Custom PCB design
- To add “Plug and Play” functionality
- To make the circuit compact

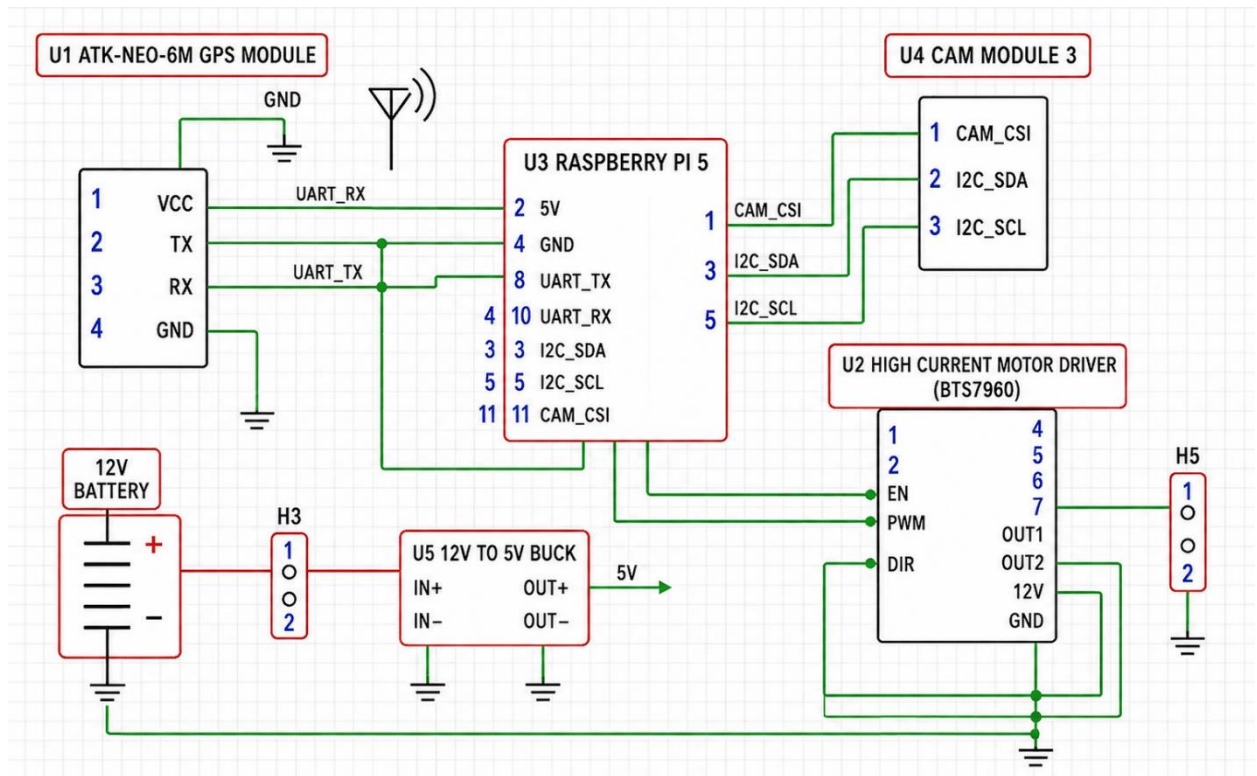


Fig (10) Tentative schematic of the custom PCB that includes GPS module & motor driver

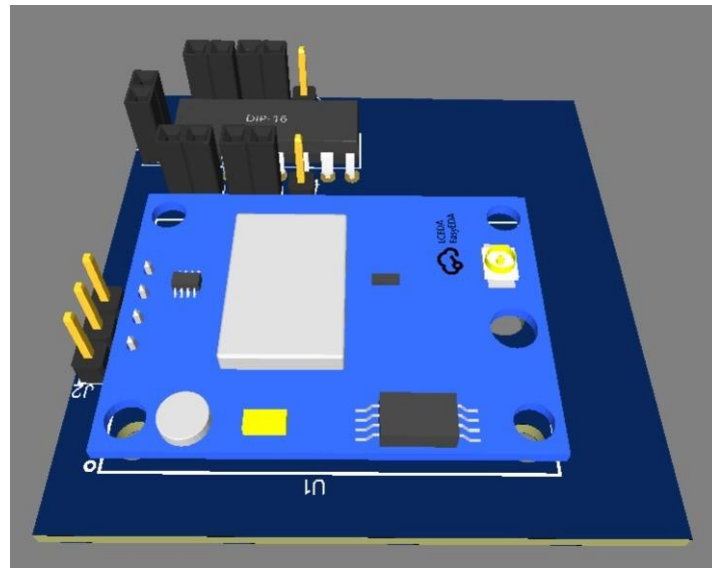


Fig (11) Tentative location of GPS module & motor driver

2.2.2 Algorithm

- YOLOv8n / YOLO11n inspires the object detection algorithm.
- A compact neural network will be used so that it can run in real time on Raspberry Pi 5.

- The libraries that will be utilised for computer vision and machine learning are OpenCV, Picamera2 and a lightweight YOLO runtime (ONNX / NCNN).
- QR codes and AprilTags will be read with OpenCV. Raspberry Pi 5 is chosen as the onboard computer because Robofest 6.0 allows Raspberry Pi or Jetson Nano, and this stack is sufficient for navigation, sign detection, obstacle cues and decision making.

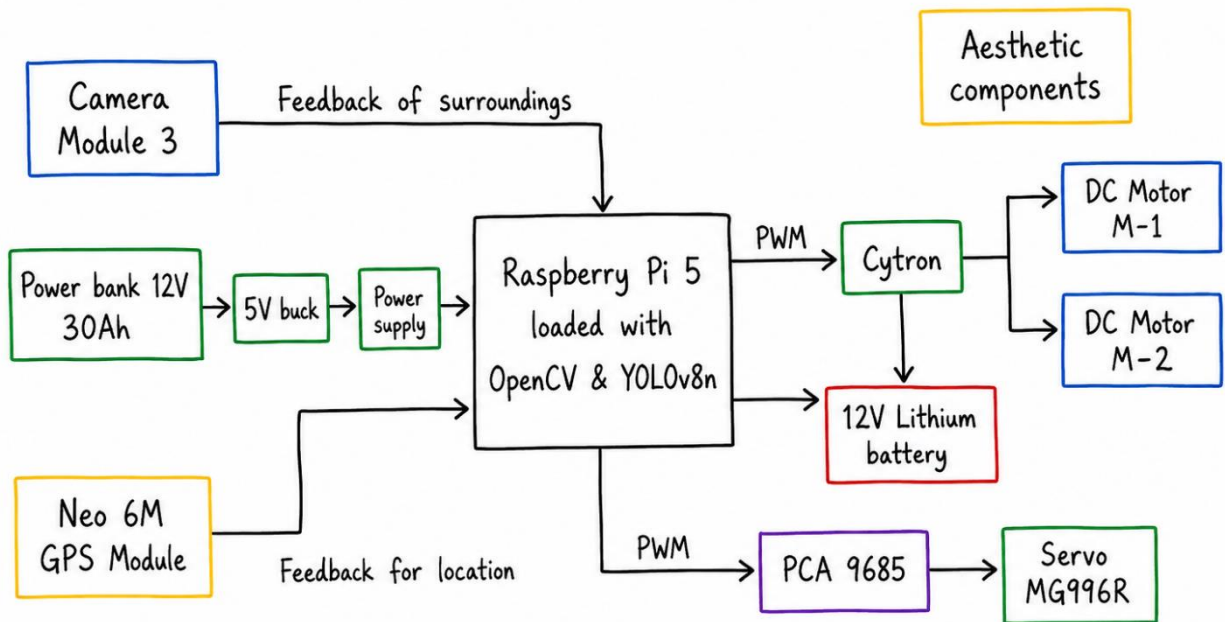


Fig (12) Control signal diagram of the circuit

3. Design

This section develops an intuitive overview of the structural aspects of the proposed Rover.

3.1 Methodology of assembling

3.1.1 Mechanical structure assemble

- The rover's body comprises aluminium extrusions (2cm x 2cm cross-section) of needed dimensions.
- Some aluminium extrusions are welded together, while others are joined using L-shaped brackets.
- For the screwing purposes, we've used the M4 types of nuts & bolts.
- For the facilitation of smooth rotations around the rocker-bogie mechanisms, we've used the pillow block bearings at all the rotation points.
- The axles are made up of steel carbon rods.

- The steering mechanism is a gear-coupled MG996R servo motor. This will help the rover to move in any direction (refer to Fig (7)).
- The shock-ups are attached to the extrusions made using CNC models to transfer a shock directly to something as sturdy as metal.
- The camera is mounted on a pedestal to ensure sufficient ground clearance, allowing accurate image capture from the Raspberry Pi Camera Module 3. Ultrasonic sensors are mounted at the front for close-range distance.
- The entire circuitry is enclosed in a casing made of acrylic sheets to ensure the protection and organisation of components.
- This model is under iterative adjustments based on the experiment's results to refine the rover's capabilities and ensure smooth operation.

3.1.2 Electronics & circuits

- Raspberry Pi 5 is supposed to be powered at 5V through a buck converter from a battery of 12V, 30Ah.
- We may need an extra power path for driving the motors, as 6 motors draw a considerable amount of current. As we want the battery to function for as long as possible, we may consider having two individual power sources: 12V for motors and 5V for Raspberry Pi 5.
- Another component that will utilize some power will be the GPS module, Neo-6M. As it does not need more than 5V, we can run it on the Raspberry Pi 5 5V rail.
- The rover is supposed to move from one specified point to another while avoiding obstacles. So, the rover will receive control signals from the GPS module, Camera Module 3 and ultrasonic sensors.
- Thus, the code is written such that the rover stays on a specified path (with the coordinates from Neo-6M) and moves such that the obstacles are avoided locally with the help of the camera and ultrasonic input.
- The servo motors require a PWM signal to function. So we've used an I²C-bus controlled 16-channel PWM driver that helps us to manage many PWMs together, a PCA 9685.
- All the necessary components are compiled over one PCB to make the rover a simple plug-and-play model.

3.2 Description of Materials

3.2.1 Components for the Mechanical body

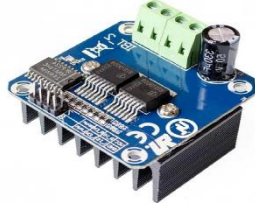
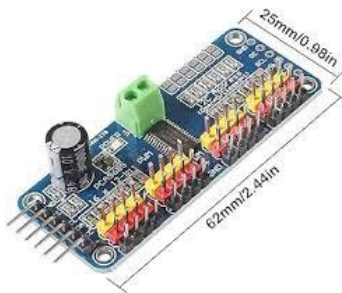
Components and Materials	Application	Specification
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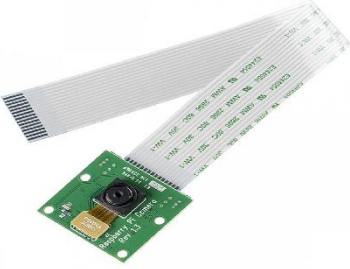
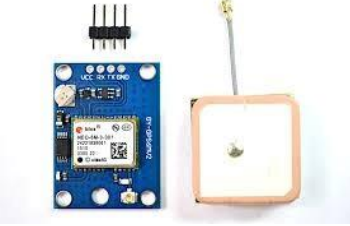
Aluminium extrusions	used for constructing lightweight, durable chassis frames	2cm × 2cm cross-section
Slider square-fitted nuts	Provide adjustable and secure attachment points to the extrusions	M4 type sliders
Servo Horns	Transfer torque from servo motors to mechanical linkages to the tires.	Metallic (aluminium)
L-shaped extrusion connector	Support modular assembly, offering stability and easy connection between extrusions.	Aluminium casted L shaped connector with 4 holes of M4 type
Flange Bearings	For smooth motion and structural support to the servo motor	10 mm diameter
Pillow Bearing	For smooth motion and structural support to the extrusion for free rolling	10 mm diameter
Steel carbon rod	Attach between the bearing and Hold the load of the chassis	10 mm diameter
Bolts	Bolts secure components and structures in rover assembly	M4 type
3D print	Supports for servos and motors and for connectors in the chassis	ABS & PLA

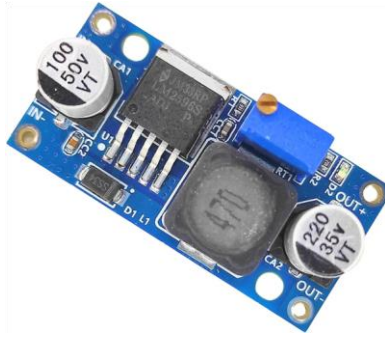
Aluminium sheets	For the motor mounting and used for the 150-degree angle connection of the extrusion	3mm thickness
Acrylic sheet	For Electronics enclosure and final touch	5 mm thickness
Camera pedestal / mount	Holds Raspberry Pi Camera Module 3 at height	3D-printed ABS/PLA or aluminium
Ultrasonic brackets	Front distance sensors	3D-printed ABS/PLA

Note:- The list of mechanical elements is subject to change. We may or may not use the elements listed above or add any if necessary; however, the most probable ones are listed above. Any sort of change will be documented in the repository (link on pg 21)

3.2.2 Electrical Component Description

Components and Materials	Application	Images
BTS7960 / Cytron-class high-current motor driver	To precisely control the 6 drive motors	
PCA 9685	Controlling pulse width modulation (PWM)	

Raspberry Pi Camera Module 3	Capture RGB data for navigation, signs, QR and parking	
Ultrasonic sensors	Close-range distance for obstacles, tunnel and underpass	
Raspberry Pi 5 Board (8 GB)	Computer for processing data and decision-making	
Neo-6M GPS Module	To aid in navigation using GPS	
Batteries	Power supply (12V, 30Ah)	

Buck Converter	Step-down 12V to 5V DC for Raspberry Pi 5	
Servo motor	Wheel steering (MG996R Servo)	
DC Motors • 130 RPM speed • 38 kg-cm torque	Locomotion of the rover, i.e. for providing speed and torque.	

3.3 Rough dimensions proposed for the proof of concept using CAD

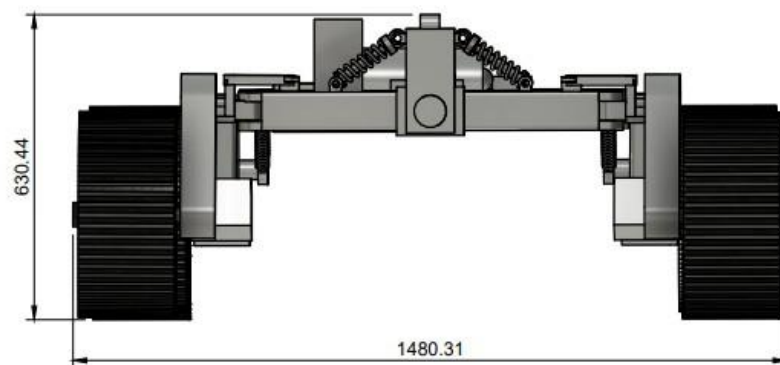


Fig (13) Front view

Note: Figures 13-16 are conceptual CAD drawings from an earlier design pass; the intended build size is the one specified in Fig 17: 100 cm x 90 cm x 63 cm.

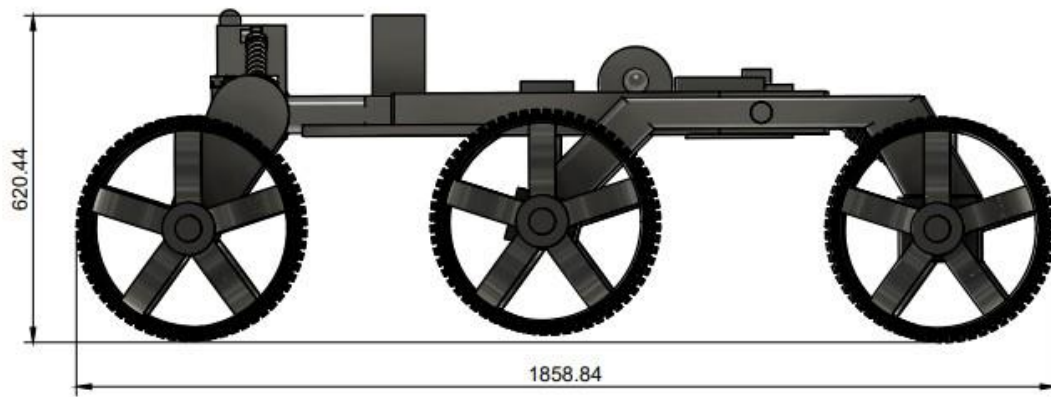


Fig (14) Side view

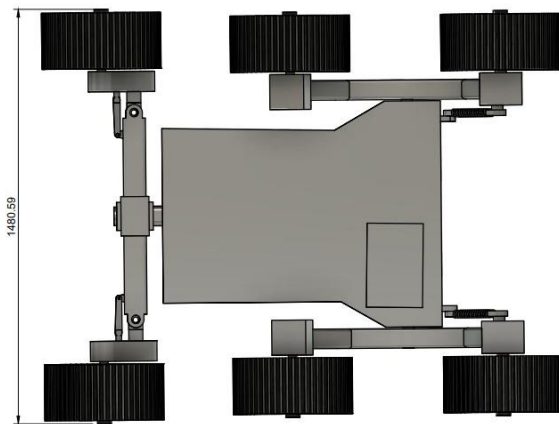


Fig (15) Bottom view

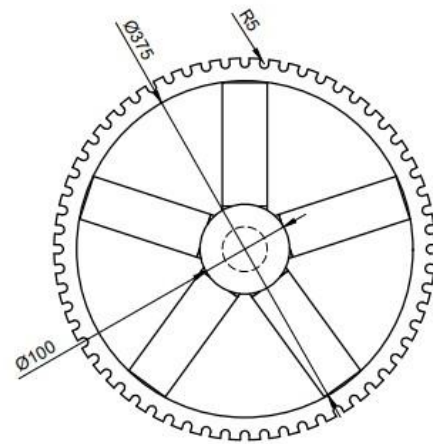


Fig (16) Wheel

3.4 Dimensions of the Model Proposed as Prototype (Actual Version)

TOP VIEW

- 1) overall length of the rover is 100 cm
- 2) overall breadth of the rover is 90 cm
- 3) Diameter of tyre is 25 cm
- 4) distance between end point of tyre are 10 and 15 cm
- 5) Breadth of tyre is 10 cm
- 6) connecting joint of axl made with 2 pillow bearing and rod for free

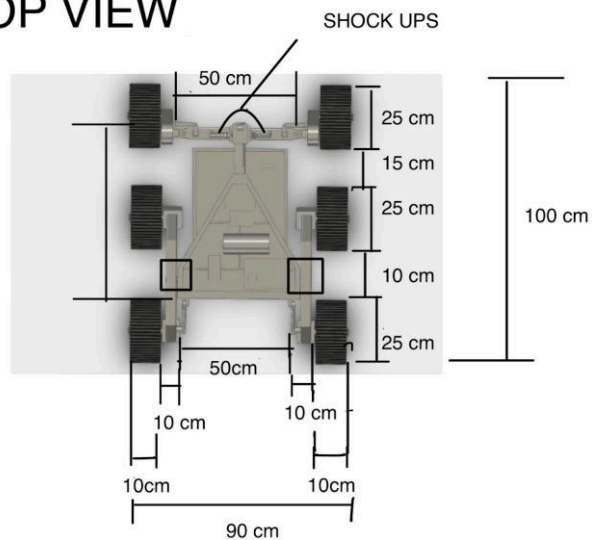


Fig (17) Top view of Rover

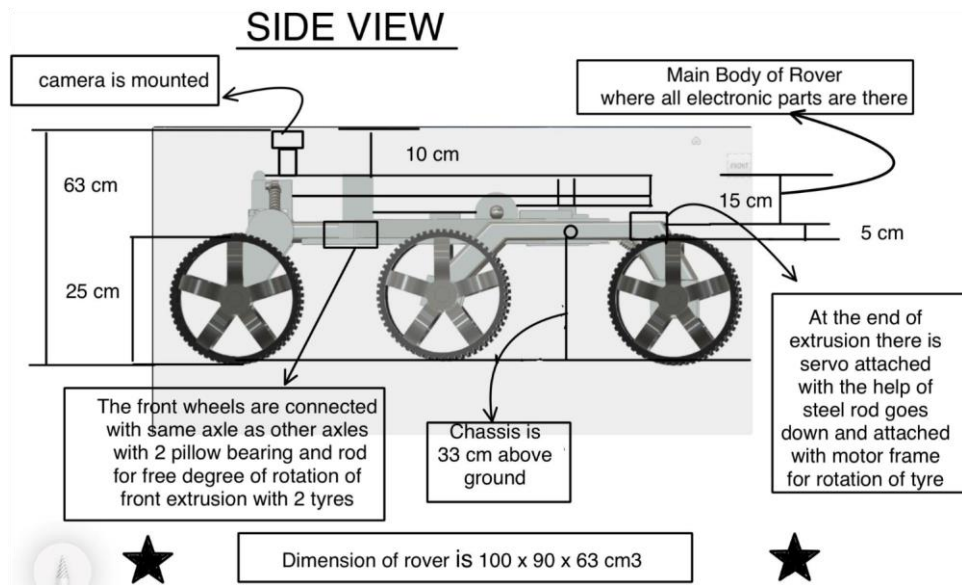


Fig (18) Side view of Rover

4 . What's running on Raspberry Pi 5?

Raspberry Pi 5 (8 GB) is a compact yet capable computer that we propose to use for running our object detection algorithms and other decision-making processes.

It has the following features that favour our use:

1. onboard computer officially allowed in Robofest 6.0
2. 8 GB RAM for camera, planning and lightweight AI models
3. native CSI camera support for Camera Module 3
4. CPU suitable for real-time OpenCV and YOLOv8n / YOLO11n
5. compact board size for mounting on the rover
6. 40 GPIO pins for additional peripherals and devices

It also provides a Python library for directly accessing the GPIO pins. Below is a sample program for detecting an event from GPIO pin 4 and giving output to pin 18.

```
from gpiozero import DigitalOutputDevice, Button
from time import sleep

out_channel = DigitalOutputDevice(18) # output pin
in_channel = Button(4)                # input pin

out_channel.on()    # start HIGH
state = 0

try:
    while True:
        if in_channel.is_pressed:
            if state:
                out_channel.on()
                state = 0
            else:
                out_channel.off()
                state = 1
        sleep(0.2)
except KeyboardInterrupt:
    out_channel.close()
```

4.1 Libraries

1. NumPy: For efficient numerical computation
2. SciPy: For scientific computation
3. Matplotlib: For creating interactive visualisations
4. Picamera2: Python library for Raspberry Pi Camera Module 3
5. OpenCV: Computer vision library
6. gpiozero: GPIO control on Raspberry Pi 5

4.2 Pseudo Codes

All the codes mentioned in this section are sample codes to demonstrate basic functionality.

1. DC Motor drive

- This code is a sample code for switching on & off a pin on the board.
- The code demonstrates digital signals (HIGH or LOW, nothing else).
- To inculcate the in-between values (analogue output), we can change the frequency & duty cycle of the input wave with appropriate delays (using `time.sleep(XYZ)`).

```
from gpiozero import
DigitalOutputDevice
from time import sleep

out = DigitalOutputDevice(16)  #
output pin

try:
    while True:
        out.on()      # HIGH
        sleep(2)
        out.off()     # LOW
        sleep(2)
except KeyboardInterrupt:
    out.close()
```

Fig(19) On-off of 16th GPIO Pin with a time period of 4 sec

2 . Servo motor using PCA 9685

- PCA9685 driver
- Can direct **16 PWM signals** into different slots.
- Thus, we can choose to operate **16 different servo motors** on a maximum of one PCA9685.
- Although we may not need as many as 16, discussing how to handle 16 or more is essential.
- The code below shows how to operate the servo connected to the 1st position on PCA9685. Using the exact mechanism, we can drive the remaining motors (change index 0 to any number between 0 and 15).
- If one needs to handle more than 16 motors, two PCA9685 boards can be used together.
- Raspberry Pi 5 talks to PCA9685 over **I2C**. Enable I2C on Pi 5 before running this code.

```
from adafruit_servokit import
ServoKit
import time

pca = ServoKit(channels=16)

pca.servo[0].angle = 0

for i in range(180):
    pca.servo[0].angle = i
    time.sleep(0.1)

for i in range(180, 0, -1):
    pca.servo[0].angle = i
    time.sleep(0.1)
```

Fig (20) Sweeping a servo attached to the 1st channel of PCA9685 using the PWM from Raspberry Pi 5

3. GPS Module

- The GPS module can provide much information regarding the rover's position.
- Out of that whole chunk of information, **we need to retrieve only latitude & longitude coordinates**.
- The following code is for the same. On Raspberry Pi 5, Neo-6M is connected to UART (/dev/serial0).

```
import serial
import pynmea2

port = "/dev/serial0"
s = serial.Serial(port, baudrate=9600, timeout=0.8)

while True:
    data = s.readline().decode("utf-8", errors="ignore")
    if data.startswith("$GPGLL") or
data.startswith("$GNGLL"):
        location = pynmea2.parse(data)
        latitude = location.latitude
        longitude = location.longitude
        print("Latitude:", latitude, "\tLongitude:", longitude)
```

Fig(21) Retrieving coordinates from the GPS module

4 . Data acquisition from Raspberry Pi Camera Module 3

- Raspberry Pi Camera Module 3 is a camera that captures colour (RGB) images for navigation, signs, QR codes and parking.
- As we are recording a video, we need to get each frame and then extract RGB data from it & pass for further analysis.
- Close-range distance is handled by ultrasonic sensors (this replaces RealSense depth).
- The code below extracts RGB frames and displays them on the screen.

```
from picamera2 import Picamera2
import cv2

cam = Picamera2()
config = cam.create_preview_configuration(
    main={"size": (640, 480), "format": "RGB888"}
)
cam.configure(config)
cam.start()

while True:
    frame = cam.capture_array() # object corresponding to each frame
```

```

# Convert RGB to BGR for OpenCV display
color_img = cv2.cvtColor(frame, cv2.COLOR_RGB2BGR)

# Showing the image on the screen
cv2.imshow("RGB image", color_img)

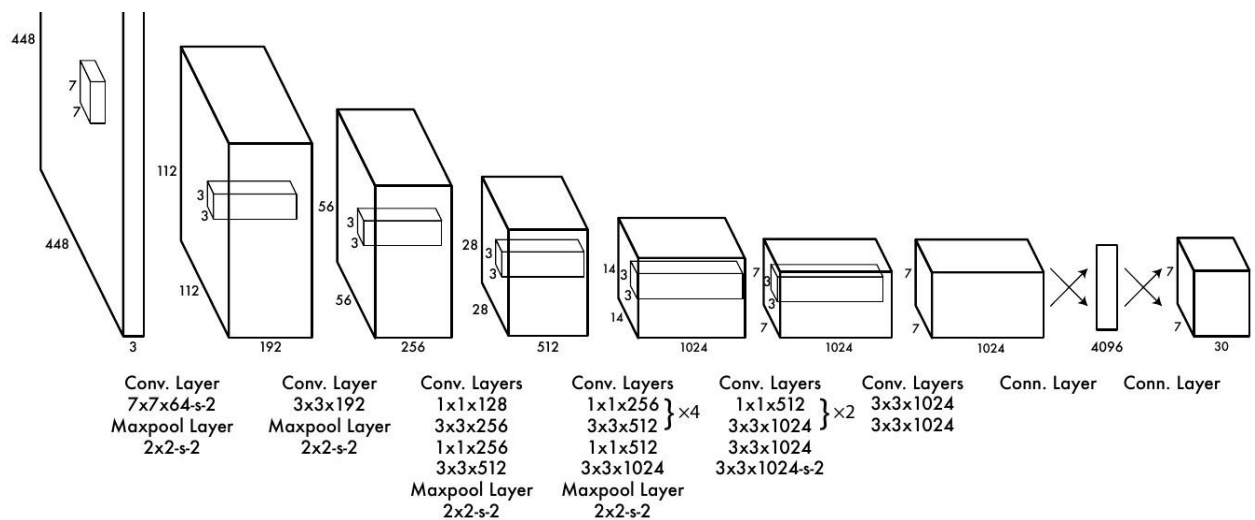
# Quitting the program
if cv2.waitKey(1) & 0xFF == ord('q'):
    break

cam.stop()
cv2.destroyAllWindows()

```

4.3 YOLOv8n

YOLOv8n is a lightweight object-detection model from the YOLO family. It uses an efficient convolutional neural network and is designed to run in real time on edge devices such as Raspberry Pi 5.



Typical Neural Network architecture of YOLO [3]



Example of Object detection using YOLO [4]

The significant advantages of YOLOv8n over other models are the ease of use, fast inference speed, and the ability to detect multiple objects in a single frame with confidence values.

5. Applications

- Self-guided navigation & AI abilities allow the rover to traverse various landscapes, enabling exploration beyond human reach.
- Data collection ability in the form of digital & image-storing capabilities allows it to expand the domains of space exploration.
- It can be used for mountain climbing and transport supplies in the camps
- Rover can be used for surveillance and monitoring purposes in sensitive or high-risk areas. It can patrol perimeters, monitor activity, and detect intrusions, enhancing security measures in various environments.
- In disaster-stricken areas, the rover can assist in search and rescue operations by navigating through debris and hazardous terrain to locate survivors. It can also deliver essential supplies and equipment to inaccessible locations.
- Potential places of use are:-
 - Space explorations
 - Rescue missions in case of calamities & disasters
 - Sample collection from challenging terrains

- Military uses such as delivery bots & patrolling
- Construction sites
- Agricultural uses such as monitoring, plantation & harvesting
- Mobile transport systems in internal spaces such as offices & townships

Description

The proposed rover can be used in several outdoor and industrial settings. It is suitable for environment checks, disaster-support surveys, farm inspection and campus or site patrol. Because it can move on mixed terrain and carry a camera plus a gripper, it can help with field observation and basic object pickup. The same platform can also support student research in autonomous navigation and practical problem-solving.

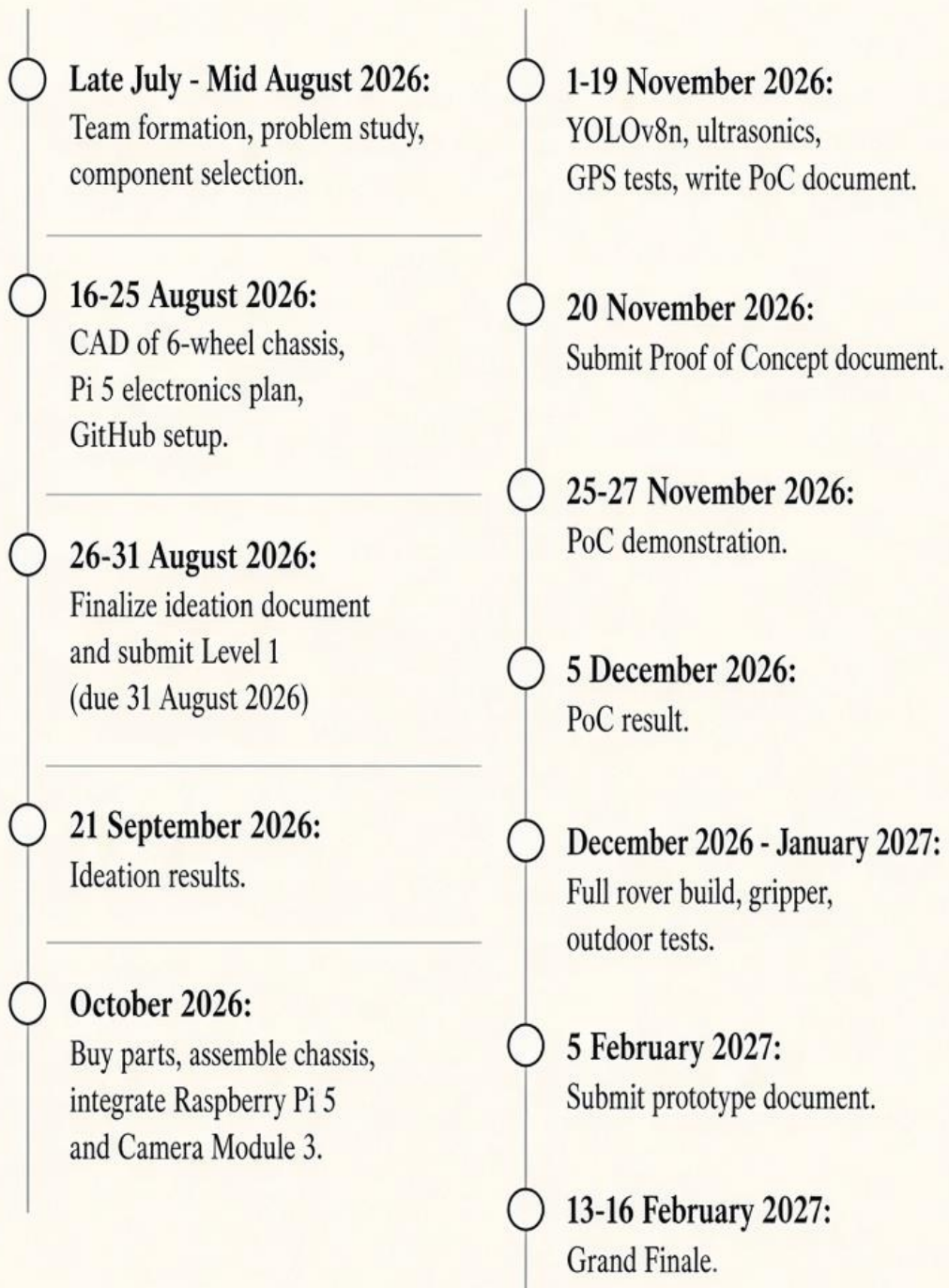
6 . Repository

GitHub repository: <https://github.com/akshpatel9604/Robofest-6.0-IGVC>

- The repository will store all essential files, including CAD screenshots, 3D-print parts, Raspberry Pi 5 code, object-detection algorithms, circuit diagrams and other required materials.
 - This also allows the team to collaborate efficiently.
 - This repository will be maintained throughout the project.
-

Timeline of Our Project

Timeline of Our Project



7 . References

- [1] Raspberry Pi Ltd, “GPIO — gpiozero,” *gpiozero.readthedocs.io*.
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